

LegalMind: An Intelligent Solution for Legal Document Analysis with User-Centric UI and AI-Driven Capabilities in Mobile Devices

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Abstract

Navigating legal documents is challenging due to their complexity, jargon, and interconnected entities like names, dates, and provisions, leading to inefficiencies and critical oversights. With the growing reliance on mobile devices, there is an increasing demand for tools that enable efficient and accessible legal document analysis on-the-go. To address these issues, we introduce LegalMind, a mobile app that revolutionizes legal document analysis by integrating key features: Automatic Entity Mapping for quick identification of essential details, Intelligent Question Answering for context-aware responses, and Multi-document Analysis for comprehensive comparison. Designed with a user-centric UI, LegalMind streamlines information retrieval, reduces manual effort, and enhances decision-making for legal professionals. The intuitive UI allows users to easily navigate complex legal content on mobile devices. Our user study confirmed that LegalMind improves efficiency and accessibility, thus showing its ability to transform legal workflows by bridging the gap between complex legal challenges and AI-driven solutions while maintaining a seamless, accessible UI.

CCS Concepts

• **Human-centered computing** → **User interface design**; *User centered design*; **Empirical studies in interaction design**; **Empirical studies in accessibility**; **Mobile devices**.

Keywords

Legal document analysis, Entity mapping, Question answering, Mobile LegalTech, AI-assisted legal workflow

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1 Introduction

Legal work is crucial across sectors, from government and corporate to personal matters, with professionals generating, analyzing, and managing extensive documents, as most legal procedures rely on creating or referencing such records [6, 25]. However, the complexity of legal language and the unstructured nature of these documents make it challenging to locate key information like case laws, legislation, names, and dates [10]. Given its reliance on language, logic, and intricate conceptual interactions, law offers a unique opportunity for computational solutions [18, 25]. Yet, the vast volume of documents and the complexity of legal language often overwhelm practitioners, hindering decision-making and increasing the risk of oversight [26]. Additionally, the need for mobile-friendly solutions is critical, as legal professionals require access to these documents on-the-go, necessitating intuitive, efficient, and responsive mobile applications that streamline document analysis and enhance usability. Advances in AI and NLP offer promising solutions by automating tasks such as document analysis, information extraction, pattern identification, and summarization, transforming the traditionally labor-intensive legal review process [11].

Despite these innovations, challenges remain—manual document review is still time-consuming, especially when searching through jargon-heavy texts [20], and traditional tools lack intuitive interfaces and multi-document analysis capabilities. This has driven the development of Retrieval-Augmented Generation (RAG) methods, which combine information retrieval and generative AI to enhance precision and contextual understanding, addressing challenges like entity mapping, context-aware question answering, and multi-document comparison, providing more efficient solutions for legal tasks.

To solve these challenges, we introduce LegalMind, an intelligent mobile app designed to address the complexities of legal document analysis by combining advanced AI technologies with Human-Computer Interaction (HCI) principles. LegalMind harnesses these innovations to optimize workflows and enhance accessibility for legal professionals. The app features *Automatic Entity-mapping*, which organizes crucial details such as names, dates, and provisions; *Intelligent Question Answering and Discussion*, delivering context-sensitive answers to queries; and *Multi-document Analysis*, enabling efficient document comparison. By incorporating RAG, LegalMind ensures precise reasoning and robust handling of legal data. Our systematic approach [14] and user-centric design showcase LegalMind's potential to alleviate workload, improve decision-making, and elevate efficiency in legal tasks.

2 System Design

As shown in Figure 1, LegalMind consists of three core components: text parsing, text processing, and the app's user interface (UI).

2.1 Text Parsing

Legal documents appear in both digital and scanned formats, each demanding distinct processing pipelines for accurate analysis. LegalMind handles this duality by integrating Optical Character Recognition (OCR) for non-digital sources and direct parsing for machine-readable files. For scanned or image-based documents—common in archives or physical filings—it uses advanced OCR systems (e.g., Tesseract fine-tuned for legal text or commercial APIs like Google Vision) with pre-processing techniques such as de-skewing, binarization, and noise reduction to improve text accuracy. Post-processing steps like spell correction and layout reconstruction ensure semantic integrity. In contrast, for natively digital documents (PDF, DOCX), LegalMind employs format-aware parsers (e.g., PDFMiner, Apache Tika) to extract structured content including headers, footnotes, and tables, while preserving contextual flow. This dual pipeline ensures robust, high-fidelity text extraction across formats, enabling consistent downstream tasks like clause extraction, citation linking, or legal summarization.

2.2 Text Processing

Text processing is a core component of LegalMind, leveraging recent advances in NLP and RAG [17] to analyze legal documents with both depth and precision. The module is structured around four interconnected processes: entity mapping, semantic understanding, contextual comprehension, and time-based summarization. Together, these ensure robust legal content analysis, facilitating faster and more reliable decision-making.

2.2.1 Entity Mapping. The foundation of any legal analysis begins with correctly identifying relevant entities such as case names, statutes, provisions, dates, and references. LegalMind integrates domain-specific Named Entity Recognition (NER) models [8, 24], trained on annotated legal corpora to ensure high fidelity in entity extraction. Unlike generic NER systems, these tools account for jurisdictional variations, legal abbreviations, and contextual nuances specific to legal writing. Preprocessing includes sentence segmentation and document structure analysis, which improve recognition of long-span and nested entities often found in contracts and case law.

Extracted entities are then mapped to standard legal taxonomies and cross-referenced with existing databases for validation. This step lays the groundwork for downstream semantic and contextual tasks by anchoring key information early in the pipeline.

2.2.2 Semantic Understanding. Legal texts often contain intricate phrasing and implied relationships that are not easily captured through surface-level parsing. To address this, LegalMind incorporates finely-tuned Large Language Models (LLMs) [7, 12] that provide deeper semantic representations of the text. These models interpret meaning beyond literal keywords—recognizing analogies, paraphrases, and legal reasoning structures across varying formats. Semantic vectors generated by LLMs help identify clause functions (e.g., obligation, exemption), connect related provisions, and detect cross-document consistency. Importantly, the model is adapted to legal discourse, ensuring precise word sense disambiguation and domain-appropriate inferences. Semantic understanding allows the system to treat the text as a network of legal ideas, rather than a collection of isolated statements.

2.2.3 Contextual Understanding. While semantic interpretation reveals underlying meaning, contextual understanding ensures that terms and arguments are interpreted within their appropriate legal framework. LegalMind's RAG-based model [23] performs context-aware reasoning by retrieving related statutes, precedents, or clauses to situate the document within a broader legal discourse. This is particularly vital for resolving ambiguities—such as distinguishing between jurisdiction-specific uses of identical phrases or assessing the legal weight of cited precedents. The model is trained to handle temporal and hierarchical context, recognizing when clauses depend on prior sections or external sources. It also uses prompts designed to simulate legal argument chains, enhancing interpretive coherence. By maintaining contextual integrity, LegalMind ensures legal accuracy and reduces misinterpretation in document summarization and analysis.

2.2.4 Time Series Information Processing. Legal documents often span extended timeframes, involving sequential events, procedural steps, or historical references. To facilitate quick comprehension, LegalMind introduces a Time Series Summary feature powered by LLMs. This functionality extracts and chronologically organizes key events, deadlines, filings, and decisions into concise narratives. The model infers temporal links even when dates are implied or scattered across paragraphs, offering users a clear linear view of a case's progression. Visual summaries or timelines can be generated to support further legal strategy or reporting. This feature improves situational awareness and allows legal professionals to grasp procedural flow without reading entire case histories.

2.3 App Features and UI

Our LegalMind app offers a suite of innovative features designed to enhance the efficiency, accessibility, and reliability of legal document analysis for both professionals and non-experts.

2.3.1 Automatic Entity-mapping. It simplifies legal document review by identifying and highlighting key entities such as names, dates, case references, and legal provisions [21] (Figure 2: Snap 1,2). Using NER [8] and a Large Language Model (LLM) [7] fine-tuned

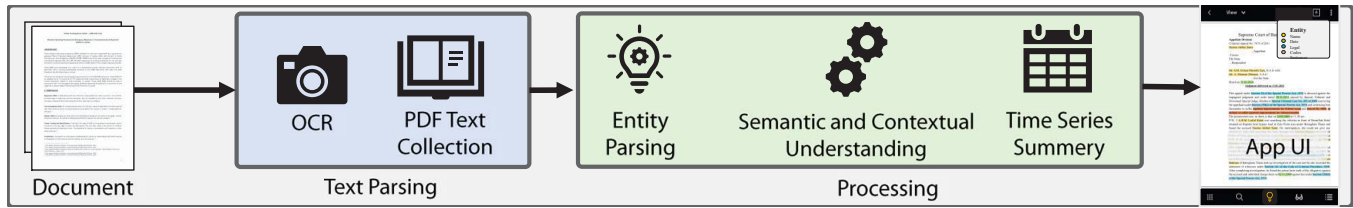


Figure 1: System Overview with different components.

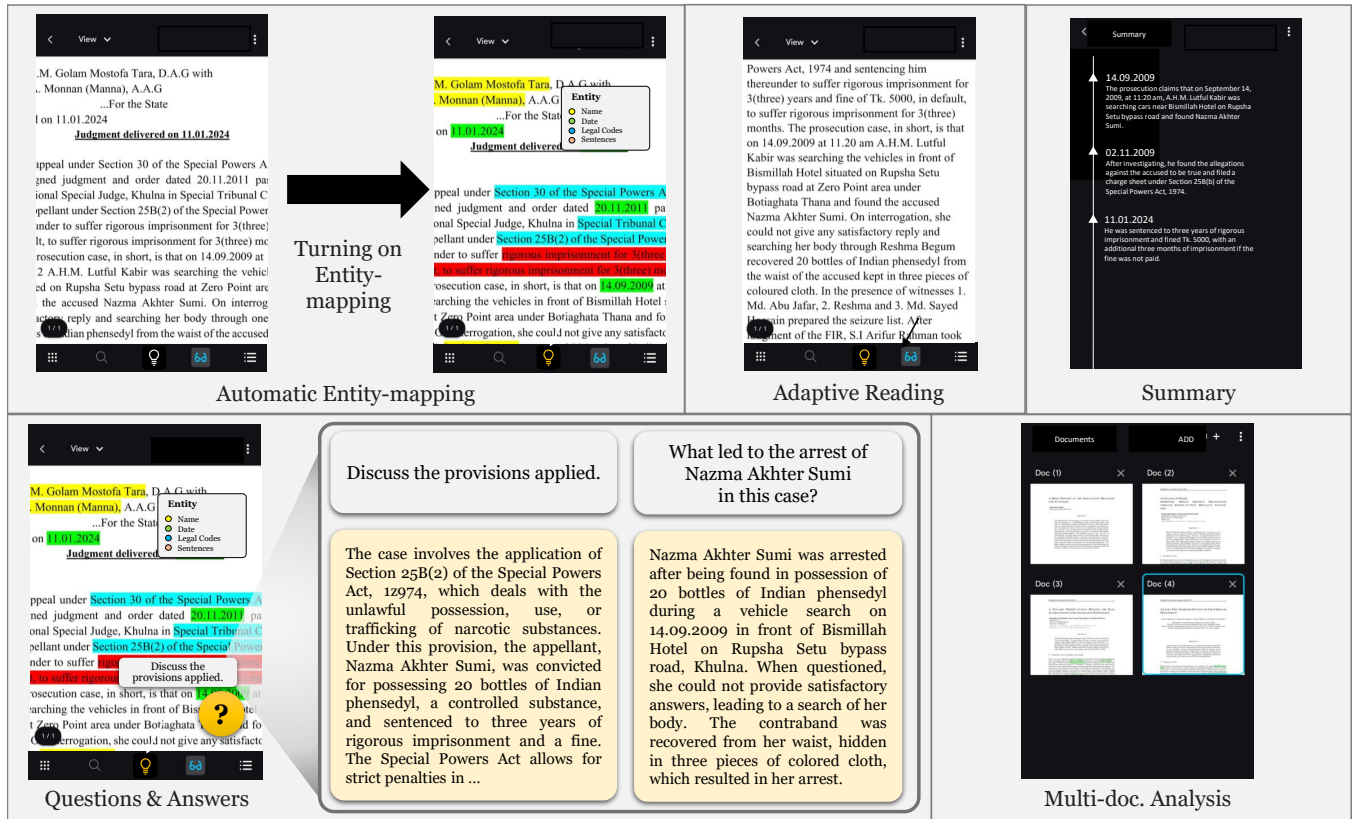


Figure 2: Demo UI Snapshots of LegalMind App.

for legal tasks, this feature categorizes and tags critical entities with increasing accuracy as it processes more documents. It solves the problem of navigating dense, jargon-filled legal texts by highlighting essential information, reducing manual effort and cognitive load [3]. Unlike traditional methods that rely on manual searches or basic keyword highlighting, LegalMind’s dynamic entity mapping offers a more efficient and accurate solution for legal professionals.

2.3.2 Adaptive Reading. This feature enhances the reading experience by adjusting the UI to be more reader-friendly [15], particularly for rigidly formatted legal documents (Figure 2: Snap 3). Legal documents often have complex layouts that are not mobile-friendly, making it difficult to read on smaller screens. This feature optimizes the layout, font size, and text spacing, making it easier for users to focus on the content. Context-aware algorithms ensure that critical

sections, such as legal clauses or key terms, remain visible and readable [22]. This addresses the problem of navigating poorly formatted legal documents on mobile devices, improving readability and user experience. Adaptive Reading allows users to interact with content in a way that suits their preferences, reducing cognitive overload and enhancing comprehension. Unlike traditional methods relying on fixed formats, this dynamic, user-centered interface improves accessibility and efficiency in legal document analysis.

2.3.3 Temporal Summary with Calendar of Events. Legal documents often embed critical events across paragraphs and pages, making it difficult for professionals to manually track procedural developments and deadlines. LegalMind addresses this challenge with an LLM-powered Temporal Summary module that automatically extracts time-bound events and organizes them into a structured,

chronological timeline (Figure 2: Snap 4). Drawing from prior work in temporal information extraction [2, 19], the system uses rule-based parsing combined with temporal tagging models to identify explicit and implicit dates, event triggers, and temporal adverbials within legal texts. Each event is normalized to a standard calendar format (ISO 8601) and anchored using temporal expressions derived from TimeML-based annotations. The module employs dependency parsing and clause segmentation to associate dates with legal events (e.g., “judgment rendered,” “contract effective,” “appeal filed”). A secondary sequence alignment process ensures the chronological ordering of related but separately mentioned events. Users can view the generated timelines through a visual interface or download them as structured summaries (e.g., CSV or JSON). This feature not only enhances comprehension but also supports deadline tracking and strategic legal planning.

2.3.4 Intelligent Question Answering and Discussion. LegalMind offers an interactive intelligent question answering (QA) and discussion interface that enables users to pose targeted queries grounded in legal documents (Figure 2: Snap 5). Powered by a domain-specific fine-tuned LLM [1], the QA engine combines document embeddings with RAG to support both extractive and abstractive answers. It retrieves relevant passages via BM25 or dense vector methods (e.g., SBERT, Contriever), then applies context-aware reasoning using LLMs trained on legal QA datasets like COLIEE and LexGLUE [13]. The system handles structured (e.g., “What are the arbitration clauses?”) and open-ended (e.g., “Summarize Party B’s responsibilities”) queries. A query classifier detects the intent and applies prompt templates to ensure formal, accurate responses. Unlike basic keyword search, this approach offers high-precision, context-sensitive insights accessible to non-experts. The discussion feature also supports follow-up questions, maintaining coherence through conversational memory.

2.3.5 Multi-document Analysis. One of LegalMind’s most powerful features is its multi-document analysis capability, enabling simultaneous review of legal texts such as contracts, agreements, or case judgments (Figure 2: Snap 6). Built on a layered architecture, it integrates cross-document entity linking, semantic alignment, and clause-level comparison using RAG pipelines [4, 9, 17]. Legal texts are embedded using domain-specific models like Legal-BERT or Law2Vec, then indexed via FAISS or Annoy for efficient nearest-neighbor retrieval. When a user submits a query (e.g., “Compare termination clauses”), the system retrieves semantically aligned sections and generates contrastive summaries using an LLM. These summaries emphasize differences in scope, obligations, jurisdiction, and penalties. Additionally, the system identifies reused vs. customized clauses via syntactic similarity and Levenshtein distance. A differential viewer presents aligned content side-by-side, highlighting semantic divergences for easier clause tracking. This streamlines legal review, reduces cognitive load, and promotes more consistent drafting.

3 User Study and Results

We conducted a comprehensive user study to assess the effectiveness and practicality of our app prototype in real-world legal scenarios. The study involved 24 participants, including 8 law students,

8 academics, and 8 legal practitioners, ensuring a diverse range of perspectives. Remarkably, 23 out of 24 participants expressed a clear need for more efficient methods to read and analyze legal documents on mobile devices, strongly validating the relevance and demand for our app. The questionnaire focused on evaluating the app’s design, accessibility, and user interface components.

Questionnaire Design: We started by presenting our motivation, contribution, app features, and demo UI examples with use cases (Figure 2) to the participants. Following this introduction, participants were asked to evaluate the app through a questionnaire based on five key aspects of successful product design, in alignment with the System Usability Scale (SUS) [5]: *Helpfulness, Ease of Use, Intention of Use, Layout Adequacy, Quality, Overall Evaluation*:

- (1) Helpfulness: I found the app helpful for working with legal documents.
- (2) Ease of Use: I found the app easy to use.
- (3) Intention of Use: I would like to use this app in future.
- (4) Layout Adequacy: I found the app and layout components are fairly simple and necessary.
- (5) Quality: I thought the core features of the app were well-designed, important and high-quality. and
- (6) Overall Evaluation: I thought this app idea was well designed.

All questions use a five-point scale, with answers ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). This evaluation method has been widely used for evaluating HCI systems and interfaces [16].

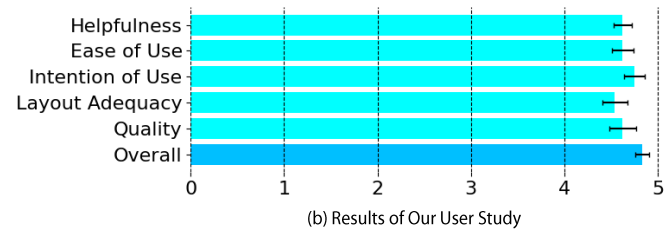
Findings: As shown in Figure 3, the participants rated our app’s overall quality very high (4.83 ± 0.38), with strong consensus on its excellence. Intention of use scored a mean of 4.75, indicating widespread interest in future use. Helpfulness, ease of use, and quality each scored 4.63, reflecting consistent satisfaction, while layout adequacy rated positively (4.54). Low standard deviations indicate uniform positive feedback. The app significantly reduced time spent locating information in legal documents and provided a smooth, intuitive experience. Even beginners navigated it easily, highlighting its user-friendliness and effectiveness.

4 Discussion

Efficiently extracting information from legal documents requires a systematic approach [8]. Law, which relies on language and logic, is increasingly suited for computational analysis [9]. Advances in artificial intelligence (AI) and Natural Language Processing (NLP) have automated tasks such as analyzing legal documents, extracting key information, identifying patterns, and summarizing content [10]. RAG is revolutionizing the legal field by drastically enhancing information accuracy and efficiency. It generates exact lines from documents and highlights risky contract clauses. It helps reduce the time it takes to read large documents, reduces human error, and speeds up automated systems. This automation reduces manual effort and time spent reviewing large document volumes, enhancing efficiency and accuracy in legal and regulatory tasks. RAG provides proper information and highlights important key points, such as dates and clauses. By making data transparent and efficient, RAG can be effectively implemented in the legal field to simplify processes. Leveraging this tool’s capabilities increases the effectiveness and accessibility of the legal industry for both professionals and commoners.



(a) Survey Participants Using Prototype



(b) Results of Our User Study

Figure 3: User study and results.

5 Concluding Remarks

LegalMind integrates AI, NLP, and Retrieval-Augmented Generation to streamline legal document analysis, automating tasks like information extraction, document comparison, and summarization. By reducing manual effort and errors, it enhances legal workflows for both professionals and non-experts. Features like Intelligent Question Answering, Temporal Summaries, and Multi-document Analysis offer precise, context-aware solutions, enabling quick access to critical information and improving accessibility. App UI is designed with a focus on usability, ensuring an intuitive and seamless experience, making complex legal content more accessible to users. User study findings confirm the app's effectiveness, with participants rating its overall quality highly and indicating strong interest in future use. While LegalMind significantly enhances legal document analysis, challenges remain. Future work will focus on refining RAG models for better interpretation of complex legal language, expanding integration with legal research platforms, and enhancing multilingual support. Adding advanced features like risk detection in contracts will further increase LegalMind's utility, contributing to more efficient and accurate legal document review and making the legal field more accessible.

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